Anthocyanin-rich grape extract blocks breast cell DNA damage.

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Anthocyanins, belonging to the flavonoid family of phytochemicals, have received attention as agents that may have potential in preventing chronic diseases such as cardiovascular diseases and certain cancers. In the present study, an anthocyanin-rich extract from Concord grapes [referred to as Concord grape extract (CGE)] and the anthocyanin delphinidin were evaluated for their capacity to inhibit DNA adduct formation due to the environmental carcinogen benzo[a]pyrene (BP) in MCF-10F cells, a noncancerous, immortalized human breast epithelial cell line. CGE at 10 and 20 microg/mL and delphinidin at 0.6 microM concentrations significantly inhibited BP-DNA adduct formation. This was associated with a significant increase in activities of the phase II detoxification enzymes glutathione S-transferase and NAD(P)H:quinone reductase 1. In addition, these grape components also suppressed reactive oxygen species (ROS) formation, but did not induce antioxidant response element-dependent transcription. Taken together, these data suggest that CGE and a component grape anthocyanin have breast cancer chemopreventive potential due in part to their capacity to block carcinogen-DNA adduct formation, modulate activities of carcinogen-metabolizing enzymes, and suppress ROS in these noncancerous human breast cells.